INORGANIC CUMULATIVE EXAM

September 16, 1995

The development of bonding theories used by inorganic chemists have required the constant interplay between theory and experiment. Each of the following experimental techniques has played an important role in this evolutionary process. Describe, as shown in the example, how each of these techniques has aided in the development of a Molecular Orbital description of bonding.

Example

Magnetic Susceptibility. Magnetic susceptibility measurements are used to determine the number of unpaired electrons in metal complexes. Early measurements suggested that the d orbitals were split into groups which depended on the ligands present, the geometry of the complex and the total number of d electrons. For example, all octahedral Co(III) complexes are diamagnetic with six electrons where as some octahedral Co(II) complexes have 1 unpaired electron and others 3 unpaired electrons. Another critically important example is O_2 which contains 2 unpaired electrons consistent with the MO bonding picture involving two degenerate antibonding pi-orbitals. The appropriate MO diagrams would strengthen and clarify the arguments and should be used when necessary.

1. Infrared Spectroscopy

2. Electronic Spectroscopy

3. Photoelectron spectroscopy

4. ESR